

# MTH 1030 Week 9 tutorial

- 1 Project 2 is due on Fri 15 May (**NEXT** Friday).
- 2 You can already do Q1 based on the material from this tutorial. Q2 and Q3 need what you learn in this week's lectures/next week's tutorial. It is a long assignment so you should already have started.
- 3 About half the qns are to be done by hand (not in Mathematica). Please submit your solutions for these either typewritten (using  $\text{\LaTeX}$  or the Word equation editor) or *neatly* handwritten and scanned properly (NOT a bad photograph: use a printer on campus to scan or a special PDF scanner app on your phone).

# ADDITIONAL QNS

11. (a) Prove that a convergent sequence is always bounded.  
(b) Suppose that  $\lim_{n \rightarrow \infty} a_n = 0$ , and that some  $a_n > 0$ . Prove that the set of all numbers  $a_n$  actually has a maximum member.

12. (a) Prove that

$$\frac{1}{n+1} < \log(n+1) - \log n < \frac{1}{n}.$$

- (b) If

$$a_n = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} - \log n,$$

show that the sequence  $\{a_n\}$  is decreasing, and that each  $a_n \geq 0$ . It follows that there is a number

$$\gamma = \lim_{n \rightarrow \infty} \left( 1 + \cdots + \frac{1}{n} - \log n \right).$$

This number, known as Euler's number, has proved to be quite refractory; it is not even known whether  $\gamma$  is rational.